

THERMAL PRINT HEAD

热敏打印头

ET48B-6360

APPROVED BY CUSTOMER 客 户 确 认 栏	
APPROVED DATE 确 认 日 期	

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FOR YOUR EXCITEMENT

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1. Description 概要

This specification is applied to the thermal print head that has heat elements by thick film process. It also includes drive ICs that operate as shift-registers, latches and switching transistors to drive heat elements.

本仕様书适用于此款采用厚膜工艺形成线型发热体电阻的热敏打印头。其内置含有移位寄存器、锁存器的 C-MOS 芯片来驱动各发热体电阻。

2.Outlines 性能特点

Item 项 目	Specification 规 格	Note 备 注
Effective print width 有效打印宽度	48mm	
Number of heat elements 发热体点数	384dots	
Heater resolution 点密度	8dots/mm	203dpi
Heat elements pitch 点间距	0.125mm	
Heater resistance 发热体阻值	$\bar{R} = 90 \Omega \pm 5\%$	
Number of data inputs 数据传送方式	1 serial input	Data In
Logic signals 逻辑输入信号	1 STROBE and 1 $\overline{LATCH}$	
Logic power voltage (VDD) 逻辑电压	3.3V	
Head print voltage (VH) 打印电压	3.6V	
Specification for Thermistor 热敏电阻规格	$R_{25}=30k\Omega \pm 5\%$, $B=3,950K \pm 3\%$	$R = R_{25}e^{B(1/T-1/T_{25})}$
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3. Maximum ratings 最大定格动作条件

Item 项 目	Symbol 符 号	Maximum Specification 最大规 格	Note 备 注
Heater energy consumption 印加能量（时间）	Eo max(t _s)	0.35 mJ/dot(2.4ms)	2.50 ms/line Ta = 25 ° C VH=4.2V
Head print voltage 打印电压	VH max	5.5V	Between Connector TPH 接入端
Logic voltage 逻辑电压	VDD max	5.5 V	
Number of heating dots simultaneously ON 最大同时工作点数	Ndot max	384dots	
Operating temperature※ 动作温度※	Ta	-5 ° C ~ +70 ° C	Detected temperature of Thermistor shall not exceed 65° C 热敏电阻的检测温度应不 超过 65° C
Storage temperature 保存温度		-40° C ~ +80° C	Non-operating 非工作状态
Operating humidity 动作湿度		10 ~ 90%RH	Non-condensing 无结露
Storage humidity 保存湿度		5 ~ 90%RH	

Note 3.1 :

These ratings aren't a condition to guarantee the printed optical density and printhead life.
在上述动作条件下，TPH 难以保证打印质量及寿命。

※ Beyond the range of operating temperature from +5° C to +40 ° C, TPH's printing quality will be affected.

※ 在动作温度 +5° C ~ +40 ° C 范围之外, TPH 打印质量会受到影响。

4. Standard printing conditions 标准打印条件 (湿度:40%-60%RH)**4.1 Mechanical Conditions** 机械参数

Item 项 目	Recommended conditions 推 荐 条 件	Unit 单 位	Note 备 注
Platen diameter 胶辊直径	Φ8 MAX.	mm	
Platen hardness 胶辊硬度	40 ~ 50	deg	
Platen pressure 胶辊压力	11.9 ~ 18.4	N / TPH	
Scanning resolution 副扫描密度	16	line/mm	400 line/inch
Thermal paper 热敏纸	PD160R OJI PAPER Co.,LTD.(王子制纸)		Note 4.1

Note 4.1 :

When the other thermal paper is used, print conditions change by thermal paper sensitivity.

如果使用其他型号热敏纸, 打印条件应该依据热敏纸的感度相应调整。

4.2 Electrical Conditions 电气参数

Item 项 目	Symbol 符 号	Recommended conditions 推 荐 条 件	Unit 单 位	Note 备 注
Print Speed 打印速度		2.5	ms/line	
Heater power consumption 发热体印加功率	Po	0.102	W/dot	$\bar{R}=90\Omega$
Head print voltage 打印电压	VH	3.6	V	Between Connectors TPH 接入端
Heater energy consumption 发热体印加能量(时间)	5°C 25°C 40°C Eo (ts)	0.19(1.86) 0.18(1.76) 0.17(1.66)	mJ/dot (ms)	$\bar{R}=90\Omega$ Note 4.2
Supply current 消耗电流	Io	33.7	mA/dot	

Note 4.2 :

Supply energy is defined by the following formula.

发热体电阻单点印加能量按下式定义:

$$E_o = I_o^2 \bar{R} t_s = \frac{(VH - V_{com})^2 \cdot \bar{R} \cdot t_s}{(\bar{R} + R_{ic})^2}$$

$$R_{ic} = 2\Omega$$

: Driver IC "ON" resistance (驱动 IC 内阻)

$$t_s$$

: Strobe pulse width (选通脉冲宽度)

$$VH$$

: Head print voltage (打印电压)

$$\bar{R}$$

: Heater average resistance (发热体阻值)

$$V_{com} = 0.5\text{ V}$$

: Common voltage drop (VH 电源压降)

4.3 Print Characteristics 印字浓度/质量

Item 项 目	SHEC Print Quality Test SHEC 测试条件	Unit 单 位	Note 备 注
Optical Density 光学浓度	1.2 Min	OD	

Note 4.3 :

Printing all black. Optical density should be measured at 10mm intervals after the starting point using QUICKDens 200(GretagMacbeth) reflection density meter or equivalent.

全黑印字状态下, 从开始打印位置起副扫描方向 10mm 处浓度测定。

采用 QUICKDens 200(GretagMacbeth) 或等同设备。

4.4 Printhead Life 使用寿命

Under above standard printing conditions,the printhead life is specified below.

在上述标准打印条件下(25℃), 热敏打印头的使用寿命按如下标准:

Item 项 目	Specification 规 格	Note 备 注
Number of pulses 脉冲个数	1.0×10^8 pulses	
Run length 走行距离	50 km	Printing duty 12.5% 印字率 12.5%

Note 4.4 :

Printhead lifetime should be measured when the resistance value of the average resistance changes more than 15% compared with its initial value.

(Excluding the failed mode(like scratch etc.)caused by foreign dust etc.)

热敏打印头寿命以平均阻值相对于初始阻值变化率达到 15%为基准判定。

(走行过程中因异物等造成基板划痕、伤等不良, 不计入判定内)

5. Electrical characteristics for driver IC(C-MOS) C-MOS 驱动 IC 电气参数

5.1 Absolute maximum ratings 绝对最大定额

Parameter 参 数	Symbol 符 号	Test conditions 测 试 条 件	Ratings 规 格 范 围	Unit 单 位
Supply voltage 电源电压	V_{DD}	Surge 电涌	-0.4 ~ 6.0	V
	V_H	Surge 电涌	0 ~ 7.0	V
Input voltage for logic 逻辑输入电压	V_{IN}		-0.5 ~ $V_{DD} + 0.5$	V
Drive Supply Current 驱动电流	I_{DOL}		70	mA

5.2 Recommended operating conditions 推荐动作条件

Parameter 参 数	Symbol 符 号	Test conditions 测 试 条 件	Recommendations 推荐值			Unit 单 位
			Min.	Typ.	Max.	
Supply voltage 电源电压	V_{DD}		3.0	3.3	5.5	V
	V_H	Supply voltage for VH	—	3.6	5.5	V
Input voltage for logic 逻辑信号电压	V_{IH}		$0.8 \times V_{DD}$	—	V_{DD}	V
	V_{IL}		0	—	$0.2 \times V_{DD}$	V
Clock frequency 时钟频率	f_{CLK}	duty 50% 占空比 50%	—	—	7	MHz

5.3 Electrical characteristics 电气特性

Parameter 参 数		Symbol 符 号	Test conditions 测 试 条 件	Ratings 规 格 范 围			Unit 单 位
				Min.	Typ.	Max.	
Input current for logic 逻辑信号 输入电流	$\overline{LATCH}$	I_{IH}	$V_{DD}=3.3V$ $V_{IH}=3.3V$	—	—	2.0	uA
	STROBE			—	—	330	uA
	CLOCK			—	—	2.0	uA
	DATA IN			—	—	1.0	uA
	$\overline{LATCH}$	I_{IL}	$V_{DD}=3.3V$ $V_{IL}=0V$	-110	—	—	uA
	STROBE			-6.0	—	—	uA
	CLOCK			-2.0	—	—	uA
	DATA IN			-1.0	—	—	uA
“L” Output voltage of drivers “L” 电平输出电压		V_{DOL}	$V_{DD}=3.0V$ $I_{DOL}=70mA$	—	0.21	0.45	V
Leak current of drivers 驱动输出漏电流		I_{LEAK}	$V_{DOH}=8V$	—	—	1.0	uA/dot
Logic supply current 逻辑电源消费电流		I_{DD}	$f_{CLK}=10MHz$	—	—	20	mA
Logic supply current (Non-Operation) 静态消费电流		I_s	DATA IN/CLOCK = GND Other logic signal open 其他逻辑信号开路	—	—	100	uA

Note 5.1: STROBE includes pull-down resistance of $50k\Omega \pm 50\%$.STROBE 信号内置 $50k\Omega \pm 50\%$ 的下拉电阻。

5.4 Switching characteristics for driver ICs. 开关特性

Parameter 参 数	Symbol 符 号	Test conditions 测试条件	Ratings 规 格 范 围			Unit 单 位
			Min.	Typ.	Max.	
Clock frequency 时钟频率	f_{CLK}	Cascade 级联			7	MHz
Clock pulse width 时钟脉冲宽度	tw(T)		37.5			ns
Data setup time 数据建立时间	tsu(D)		20			ns
Data hold time 数据保持时间	th(D)		20			ns
Latch setup time 锁存建立时间	tsu(LA)		100			ns
Latch pulse width 锁存脉冲宽度	tw(LA)		50			ns
Latch to Strobe setup time 锁存至选通建立时间	tsu(STB)		336			ns
Strobe to Latch setup time 选通至锁存建立时间	th(STB)		36			us
Clock to Data out delay time 时钟至数据输出延迟时间	td (SO)				120	ns
Strobe to driver Output delay time 选通至驱动输出延迟时间	td(DO)r				36	us
	td(DO)f				36	us

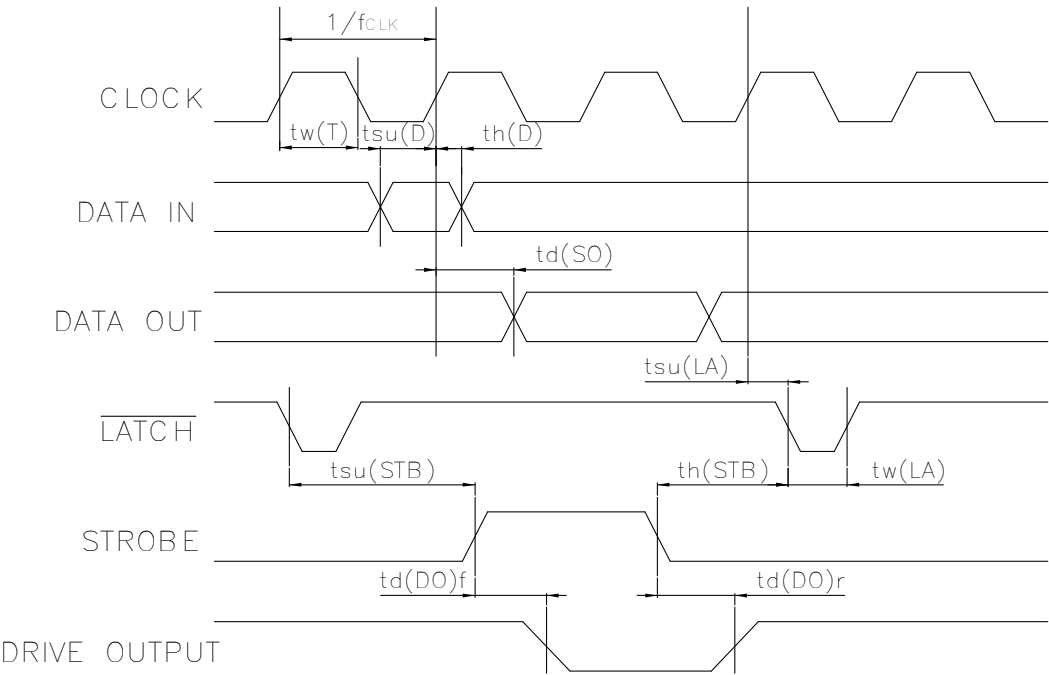


Fig. 1 Schematic Diagram 内部回路图

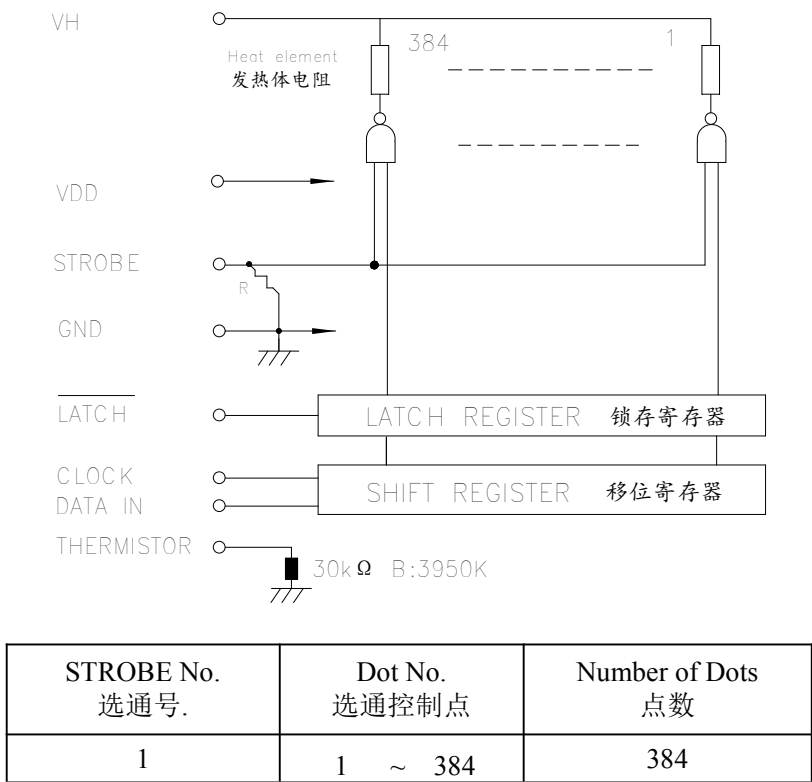
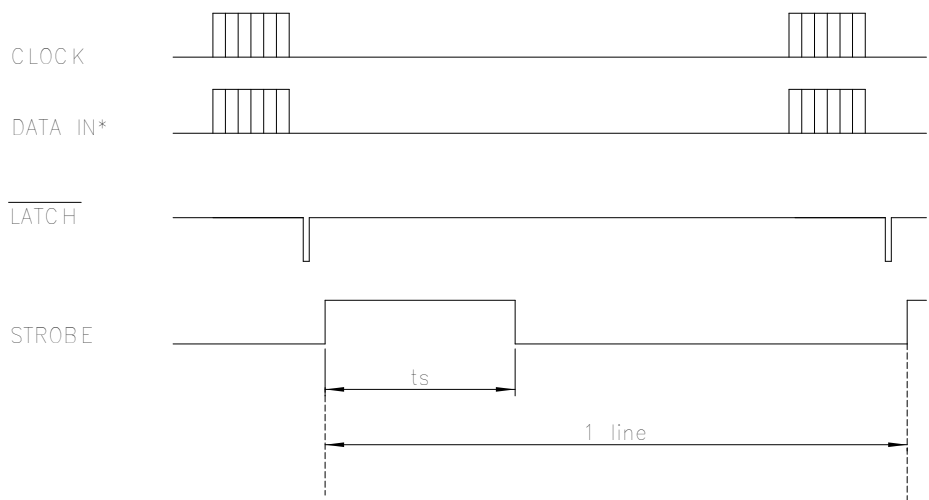


Fig. 2 Timing Chart 驱动时序图



* : Data transfer is possible in the same period as printing
在打印的同时，数据传输是可能的

Fig. 3 Cross Section (Unit:mm) 断面图(单位:毫米)

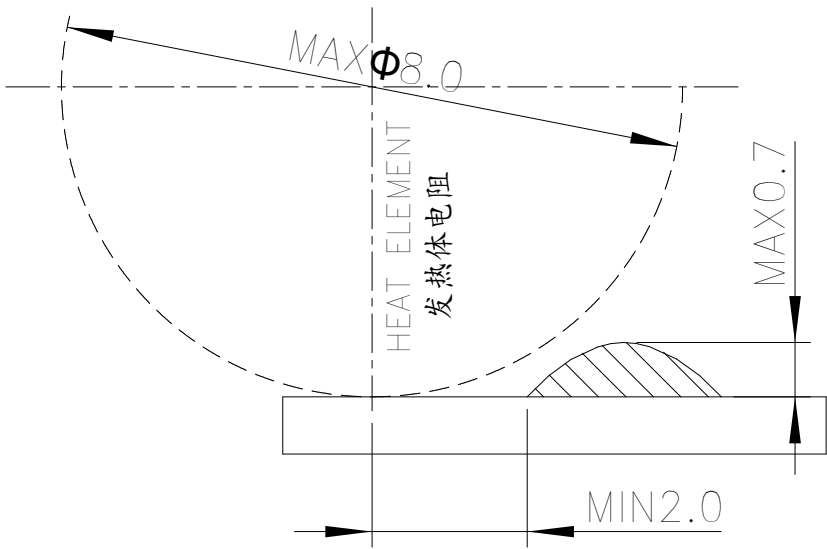
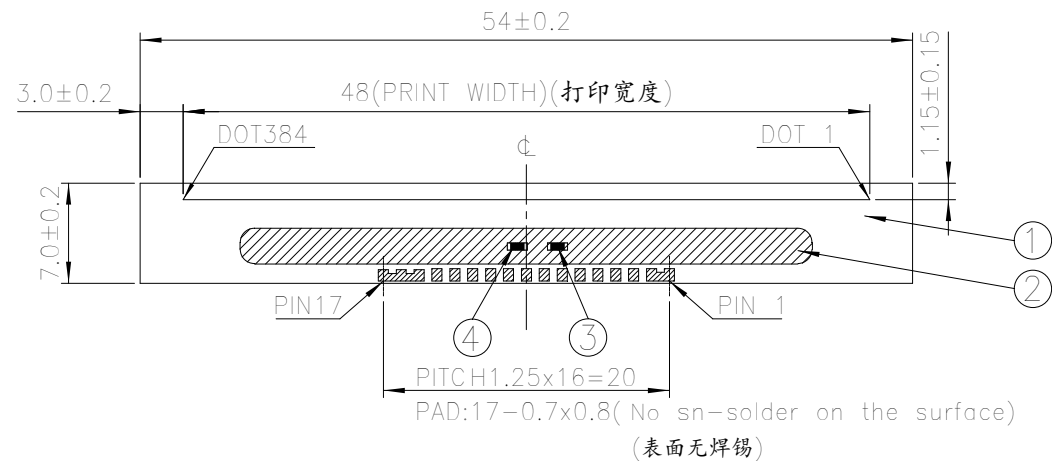


Table. 1 Pin assignment 接口定义

PIN 针脚号	NAME 信号名称
1	VH
2	VH
3	$\overline{\text{LATCH}}$
4	GND
5	GND
6	GND
7	THERMISTOR
8	STROBE
9	VDD
10	GND
11	GND
12	GND
13	CLOCK
14	DATA IN
15	VH
16	VH
17	VH

Fig. 4 Outline Dimension (Unit: mm) 外形图 (单位:毫米)



ITEM 项目	DESCRIPTION 名称	MATERIAL 材料	DEF 数量	SUMMARYS 备注
1	SUBSTRATE 陶瓷基板	CERAMIC 陶瓷	1	
2	RESIN MOLD 环氧树脂胶		○	
3	THERMISTOR 热敏电阻		1	
4	R-METAL 电阻		1	

- NOTE:
注意事项:
1. UNMARKED TOLERANCE DIMENSION IS NON-CONTROLLED
未标注公差尺寸为非管控尺寸
 2. THIS MODEL COMPLIES WITH THE RoHS DIRECTIVE.
产品所用材料均为RoHS对应品
 3. FFC/FPC TERMINALS NEED TO BE PLACED WITH Sn-SOLDER.
产品焊接的FFC/FPC端子需带锡

6. Precautions in use 使用注意事项

6.1 Electrical Precautions 电气注意

- 1) When the printer is on standby, the thermal head print voltage(VH) must be switched off.
打印设备处于待机状态(或非打印状态)时, 打印电源(VH)必须关闭。
- 2) Power supply ON/OFF sequence must be in the following order:
打印头电源须按下面的印加顺序:
Turn on : Apply the logic voltage(VDD)first, then the thermal head print voltage(VH).
打开顺序: 先逻辑电源(VDD), 后打印电源(VH)。
Turn off : Switch off the thermal head print voltage(VH)first, then turn the logic voltage(VDD) off.
关闭顺序: 先打印电源(VH), 后逻辑电源(VDD)。
- 3) During head power supply ON/OFF sequence, STROBEs signal should be kept "disable".
在打印头电源处于 ON/OFF 状态时, 选通(STROBE)信号应保持在 "disable"状态。
- 4) The voltage including surge should be kept within the Maximum ratings.
如果施加电压超过驱动 IC 的极限范围或在瞬间高电压的情况下, 锁定效应可能损坏打印头, 因此电源及信号电压(包括振荡峰值)应控制在最大规格内。
- 5) Logic signal input level is CMOS level (ex. 74HC240) or equivalent.
逻辑信号输入为 CMOS 电平(例. 74HC240 或等同品)。
- 6) The heater elements and driver ICs are electro-statically sensitive. Do not to touch directly connectors with hand or an electro-statically charged object. It is recommended that brushes near the head be provided to discharge electrostatic.
为了防止静电对驱动芯片、发热体的破坏, 注意不要用手直接接触接口端子, 也不要让端子接触带电物。建议在装配热敏打印头时, 在热敏打印头旁边装上除电刷。
- 7) Electrolytic and ceramic capacitor are recommended to be applied between VH/VDD and GND for stable power supply.
为保持电源稳定及防止瞬间高电压, 建议在打印电源(VH)和逻辑电源(VDD)与地线之间加入电解或陶瓷电容。
- 8) It is recommended that the length of cable be minimized and a flat cable with shield, cable length do not exceed 30cm.
与打印头相连的电缆长度应在 30cm 以下, 建议使用带保护层的扁平电缆。

6.2 Mechanical Precautions 机械注意

- 1) The print head substrate should be avoided to stress or shock (including dust scratch damage). And on the substrate surface near the heater, do not apply any hard material. The abrasion resistant layer is fragile to mechanical impact.
打印头基板表面应该避免重压和刮擦, 发热体受机械撞击时易碎裂, 因此要防止碰到任何硬物。
- 2) Condensation should be avoided. If condensation occurred, do not switch on the printhead power until condensation disappeared.
避免打印头表面出现结露(冷凝)。如果发生结露时, 不要开启打印头的电源,直至结露消失。
- 3) The thermal paper does not include bad element factor to affect the print head life.
避免使用表面粗糙的劣质纸张或其他介质, 否则会降低打印头寿命。在使用之前, 请确认评估热敏介质的质量。
- 4) The change of print head flatness(warp) is minimized with temperature change in the free body of one unit. Please pay attention to the warp created by fixing the printhead to the printer unit with screws or clamps.
打印头单体的平整度随温度的变化已被控制在最低程度, 当安装固定打印头时, 请避免发生扭曲或变形。
- 5) Printing without thermal paper should not be done, it may cause destruction on the print head due to overload.
应避免发生无纸打印的现象, 否则会损坏发热体电阻。
- 6) Thermal print head is heat-generating device, so that it may cause heat element damage or smoke / fire on its components by over heating, if there occurs mechanical or electrical abnormalities. So be sure to perform temperature control by thermistor and also to design the power (VH,VDD) shutdown system when abnormalities take place to protect the print head.
热敏打印头是一种发热装置, 因此如果有机械或电气方面的异常状况发生时, 打印头可能会因为过度发热而造成发

热体破坏或发烟起火。因此请确认采用打印头的热敏电阻来控制打印头温度，并设计相应的电源切断系统，异常时以便保护打印头。

- 7) Thermal paper may be affected to the print head life or caused of corrosion defect by Na⁺, K⁺ and Cl⁻ ion which are included in the media. Reliability of thermal media should be confirmed by evaluation prior to use.

Na⁺、K⁺、Cl⁻离子等会腐蚀损坏打印头，与打印头经常接触的部件，如胶辊、热敏纸等耗材请确认评估可靠性。

6.3 Handling Precautions 操作注意

- 1) When thermal paper or ink residue were stuck on the heater element area. be care to use applicator with alcohol to clean up. Sandpaper which mesh is greater than or equal to 10000# and other hard materials shall not be used to be polished .

当发热体上粘附有纸屑、灰尘或油墨残留物等时，请用柔性材料蘸酒精或洗涤剂轻轻擦拭，严禁使用粗于 10000# 砂纸等硬质材料进行研磨。

- 2) Connector pins are easy to bent, pay attention to plugging in or out.

接口插座或端子容易弯折，注意插拔时不要造成损坏。

7. Warranty 保质期

- 1) SHEC warrants this product against defects in material or workmanship for a period of twelve (12) months from the date of delivery within the limits of reliability to use.

产品的保质期为一年。从发货日起 12 个月之内，在有限责任范围内对产品的原材料或加工品质问题予以保证。

- 2) SHEC will repair or replace the product which is determined to be defective by SHEC's analysis.

经 SHEC 分析确定由上述原因造成产品不良，将对不良品进行修理或者更换。

8. FPC/FFC heat-pressing precautions FPC/FFC 热压着事宜:

In order to avoid the failures like internal-wires break caused by Resin Mold external crack or internal crack while FPC heat-pressing, the nitrogen-filling packaging for product have implemented. Anti-damp management and Assembly management also need to proceed after opening package. Please pay attentions to the following items:

为了避免 FPC 热压着时可能发生的封装胶裂、非胶裂的内部剥离引起的断线等不良，SHEC 对产品实施氮气充填包装。但也需要进行开封后的受潮管理、实装条件管理。谨请贵司注意以下几点：

- 1) The FPC/FFC heat-pressing should be proceed immediately after opening .If don't assemble FPC/FFC after opening, the desiccant is needed and the opening part should be sealed.(eg: tape sealing after collapsed).Or placed in a drying box and filled with quantitative N2 for drying protection, after more than 2 days without hot pressing, the product should be re-vacuum sealed packaging. With open-time going , the FPC/FFC heat-pressing might also be affected due to theproduct solder pads changed

产品开封后请尽早进行 FPC/FFC 热压着。若开封后不热压 FPC/FFC，请放入干燥剂，把开口部密封住（如折叠后胶带封口等）或放置于干燥箱内并充入定量 N2 进行干燥保护，超过 2 天不热压需将开封后的产品重新抽真空密封包装。如开封时间过长可能会导致焊盘表面状态变化而影响焊接质量。

- 2) In order to prevent the product from moisture , please manage the temperature and humidity during the storing in storage place.

(Recommended ≤30℃,≤60%RH relative humidity,≤4weeks Refer to MSL-2a level)

请管控产品保管场所的温度、湿度、存放期限，避免产品受潮。

(推荐:≤30℃,≤60%RH 相对湿度,≤4 周,参考 MSL-2a 等级)

- 3) Baking and Drying is recommended before FPC/FFC heat-pressing .

(Reference drying conditions: 120℃ 4Hr)

热压 FPC/FFC 前推荐对产品进行烘干、除湿（参考烘干条件：120℃ 4Hr）

- 4) The peak temperature should be strictly controlled while heat-pressing, operation should follow the verified optimal condition.

热压着时请严格控制峰值温度，按验证的最佳品质条件标准作业。